Crash It in the Simulator First
Turning an ML Signal into a Live Portfolio

A good correlation score is not a portfolio on its own. So, what can break between moving from a signal to live portfolio returns?

Jason Rosenfeld Co-founder & CEO, CrowdCent

DECENTRALIZED AI DAY NYC
Hosted by the Numerai Council of Elders

10 OCT 2026

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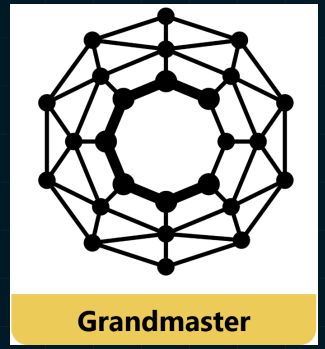
Certain slides show hypothetical, simulated performance from the CrowdCent Simulator. Hypothetical results have inherent limitations: they are prepared with the benefit of hindsight, do not reflect actual trading, and exclude slippage, market impact, and auto-deleveraging. No representation is made that any account will achieve similar results. Past performance is not a guarantee of future results.

Tournament results (Numerai, CrowdCent Challenge) are competition scores and stakes, not fund track records. Signal statistics are unaudited and subject to change. Crypto perpetual futures are highly volatile and can lose all value. Live trading features, where offered, are self-custody on Hyperliquid and are not available to US persons or residents of restricted jurisdictions.

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Numerai has proven that crowdsourced ML works. CrowdCent is among its top performers.

I've competed and staked in Numerai since 2019. CrowdCent built a hedge fund on top of Numerai to stake NMR. In 2025, we launched the CrowdCent Challenge, crowdsourcing predictions for future return rankings of every perpetual on Hyperliquid





Jason Rosenfeld

CO-FOUNDER & CEO

Ex-Millennium data scientist and investment analyst.
Numerai Grandmaster. MIDS, UC Berkeley.



Numerai Crypto, CrowdCent master model. Tournament results, not a fund track record.

The CrowdCent Challenge (Hyperliquid Ranking)

- Predict 10d and 30d relative returns for all perpetuals on Hyperliquid
- Submit every day, scored live on Hyperliquid prices
- Earn CC Points for live performance (spearman, NDCG@40, uniqueness)
- Free Data, Simulator, Trading (if in the right country), Cloud GPUs, AI Agent

70+
ACTIVE PARTICIPANTS

30K+
SUBMISSIONS

USDC + NMR
REWARDS PAID FOR LIVE PERFORMANCE, NEVER BACKTESTS

Enter with five lines of Python. Or ask your agent nicely

submit.py

```python
import crowdcent_challenge as cc
client = cc.ChallengeClient("hyperliquid-ranking")
client.download_inference_data("current", "live.parquet")
preds = model.predict(live)
client.submit_predictions(df=preds)

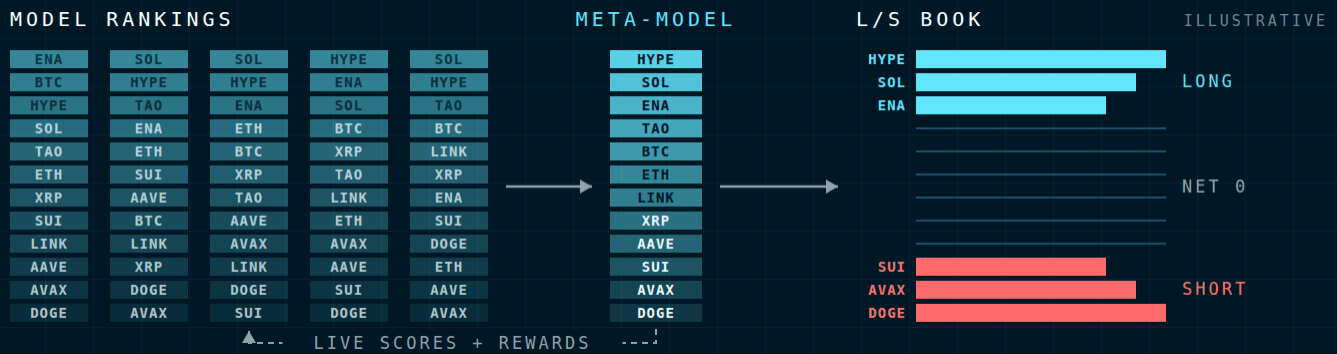
# or tell your AI agent: mcp.crowdcent.com
```

✓ SUBMITTED · SCORED LIVE ON HYPERLIQUID

Free data, a leaderboard scored on Hyperliquid prices, a Simulator that turns the meta-model into a portfolio, and CrowdCent Cloud GPUs to train on.



Many models in create one long/short portfolio.



01 Predict

Rank all ~169 Hyperliquid perps on expected 10- and 30-day relative return. New rankings every day.

02 Combine

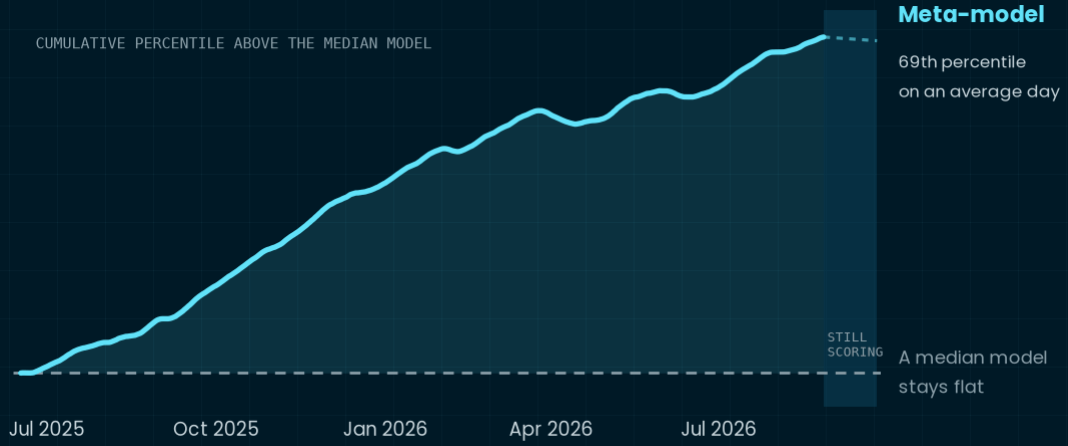
Blend the rankings into one meta-model, weighted toward models with live skill and unique signal.

03 Trade

Hold the top names long and the bottom names short, sized so long and short dollars net to zero.

WHAT GETS RANKED 169 Hyperliquid perps · 69 with over $5M open interest · ranked daily

And it works. The meta-model beats the typical model on 84% of days.



84%
DAYS ABOVE THE MEDIAN MODEL

69th
AVERAGE DAILY PERCENTILE

0.143
MEAN SPEARMAN, 30-DAY HORIZON

Every day we rank every model in the Challenge. Add up how far above the median model the meta-model lands, and the line goes up and to the right.

CrowdCent meta-model composite percentile vs all Challenge models, scored 17 Jun 2025 – 27 Aug 2026 (later days still scoring), ~180 perps. Spearman = rank correlation between predicted and realized 30-day returns.

But a live portfolio has to turn a ranking into real return. So we built a simulator.

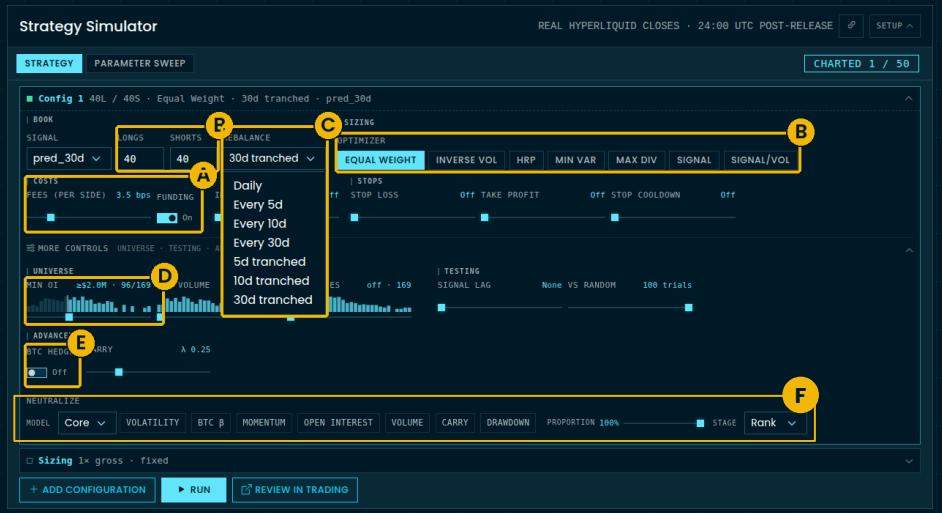
WHAT YOUR ML SCORE SEES

- Simple ranking
- Every asset counts once, liquid or not
- No fees, no funding, or costs

WHAT YOUR LIVE PORTFOLIO PAYS

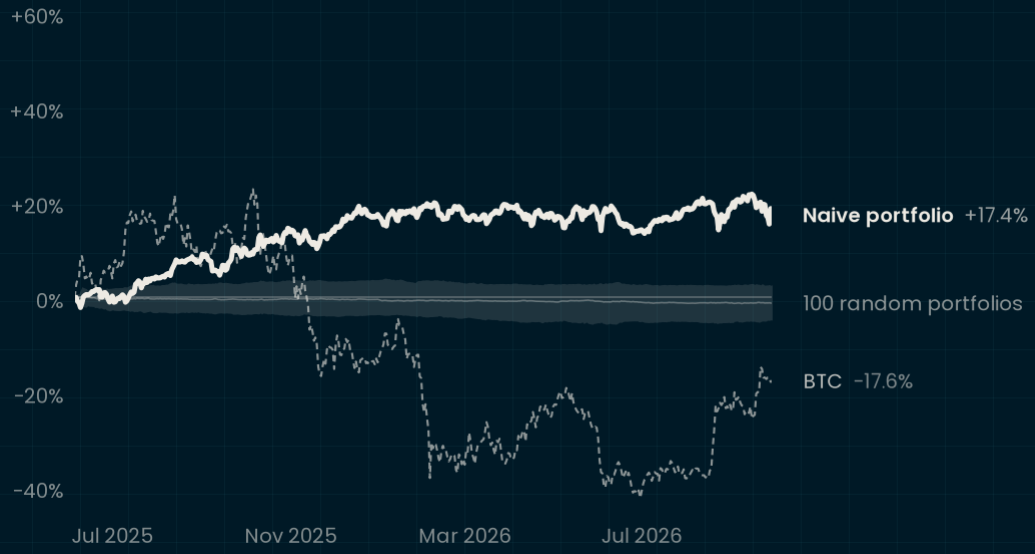
- Fees per trade, funding every hour **A**
- A few extreme names carry the P&L **B**
- Impact costs, beta, timing **D** **E** **F** **C**

MEET THE SIMULATOR LIVE ON CROWDCENT.COM



On Numerai the fund builds the portfolio for you. On CrowdCent, build your own, and test it in a simulator first.

A naive portfolio beat BTC and 100 random ones.



+17.4%
TOTAL RETURN

1.03
SHARPE

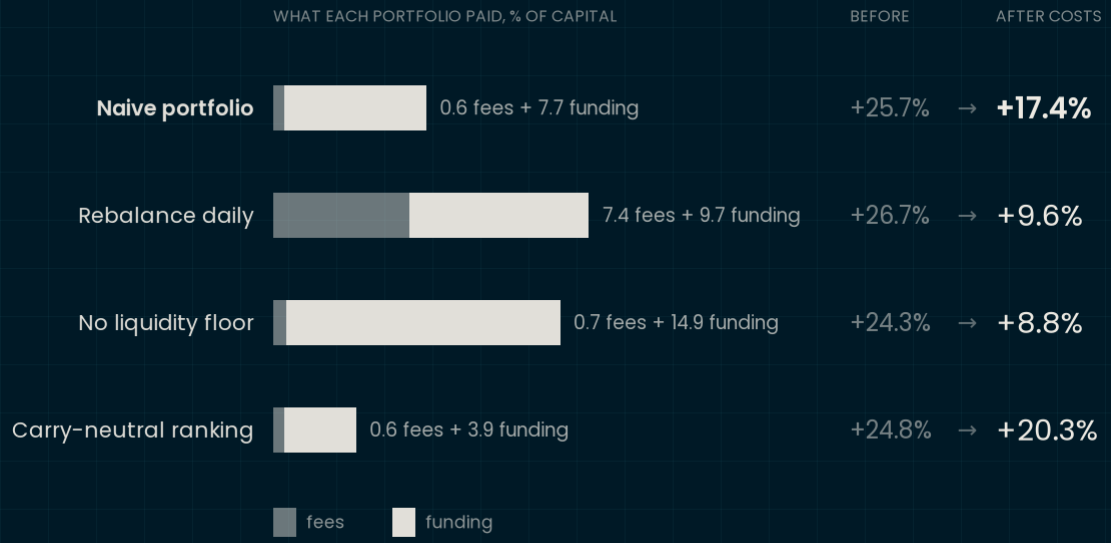
100 / 100
RANDOM PORTFOLIOS BEATEN

-5.5%
MAX DRAWDOWN

Top 30 long, bottom 30 short, equal weight, 30-day tranched rebalance, fees and funding charged, no leverage. Nothing clever, but it serves as the naive baseline

Hypothetical. CrowdCent Simulator, 30d meta-model ranking, 5 Jun 2025 – 29 Sep 2026, OI ≥ $2M, 3.5 bps fees + funding, no leverage; excludes slippage, impact, ADL. Random portfolios: 100 built from random rankings, same rules.

What costs are most important to model when converting a signal to a portfolio?

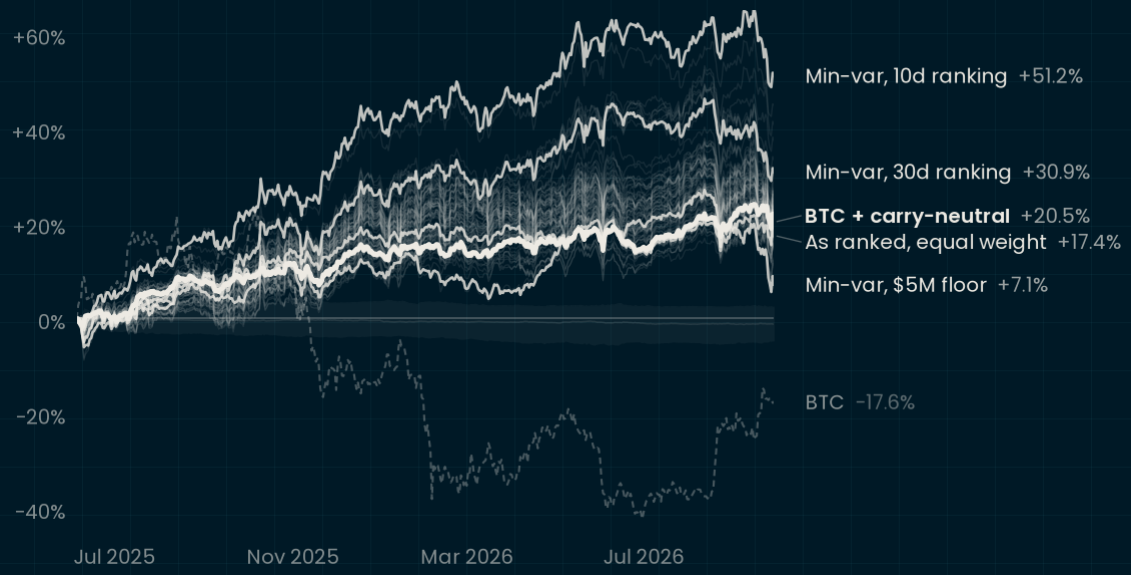


Each portfolio before and after 3.5 bps fees and funding, 30L/30S portfolio. Costs (including funding rate) can eat almost 37% of a naive portfolio's return.

A carry neutralization can reduce most of this impact. This is neutralizing to funding rate.

Hypothetical. CrowdCent Simulator, 30d meta-model ranking, 5 Jun 2025 – 29 Sep 2026. Each row moves one knob off the naive portfolio: top 30 / bottom 30, equal weight, 30-day staggered rebalance, OI ≥ $2M, 3.5 bps fees + funding, no leverage. Fees and funding paid over the period, % of capital. Carry-neutral: ranking residualized on the carry factor. Excludes slippage, impact, ADL.

Many sensible ways to build a portfolio



1.05

MEDIAN SHARPE, 80 PORTFOLIOS

+7% to +51%

RETURN RANGE, 16 MONTHS

80 portfolios: 30d ranking on 30-day tranches or 10d on 10-day tranches. Minimum liquidity filters (OI), neutral to BTC, carry or both, over various proportions.

Bold: BTC + carry-neutral, 30L/30S, equal weight, +20.5%, Sharpe 1.44.

Hypothetical. 80 CrowdCent Simulator runs on production rules: 30d ranking on 30-day staggered tranches or 10d on 10-day; liquidity floor OI ≥ $2M (30 a side) or OI ≥ $5M and volume ≥ $1M (20); as ranked or ranking neutral to BTC, carry or both; equal, signal, inverse-vol, HRP or min-variance weight (90-day risk, carry-aware, BTC constrained in the weights). 5 Jun 2025 – 29 Sep 2026, 3.5 bps + funding, no leverage. Excludes slippage, impact, ADL.

Then pick 20 that seem pretty good and net them into a single portfolio.

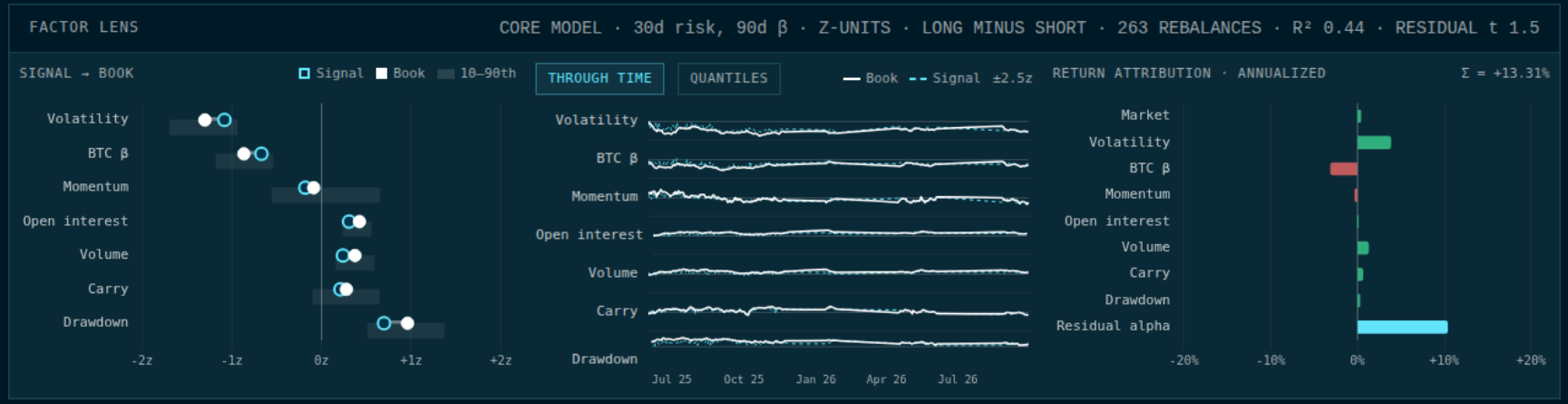


20 portfolios, netted: Sharpe 1.38, against 1.05 for the typical portfolio.

Similar return, with less volatility.

Backtest. CrowdCent Simulator, 5 Jun 2025 – 29 Sep 2026, 3.5 bps + funding, no leverage. Picked in sample, on full history: the previous slide's 80 portfolios; start from the BTC + carry-neutral portfolio, then add one at a time by lowest average correlation to those already in, to 20 (9 on 10-day). Netted in two stages: four Simulator Blends of five (the API's maximum), then an equal-weight average, which leaves out netting between groups. Excludes slippage, impact, ADL.

Decompose the portfolio’s factor exposures



SIGNAL → PORTFOLIO

BTC beta, volatility, momentum, liquidity, carry: what the signal leaned toward, and what the way you built the portfolio did to it.

RETURN ATTRIBUTION

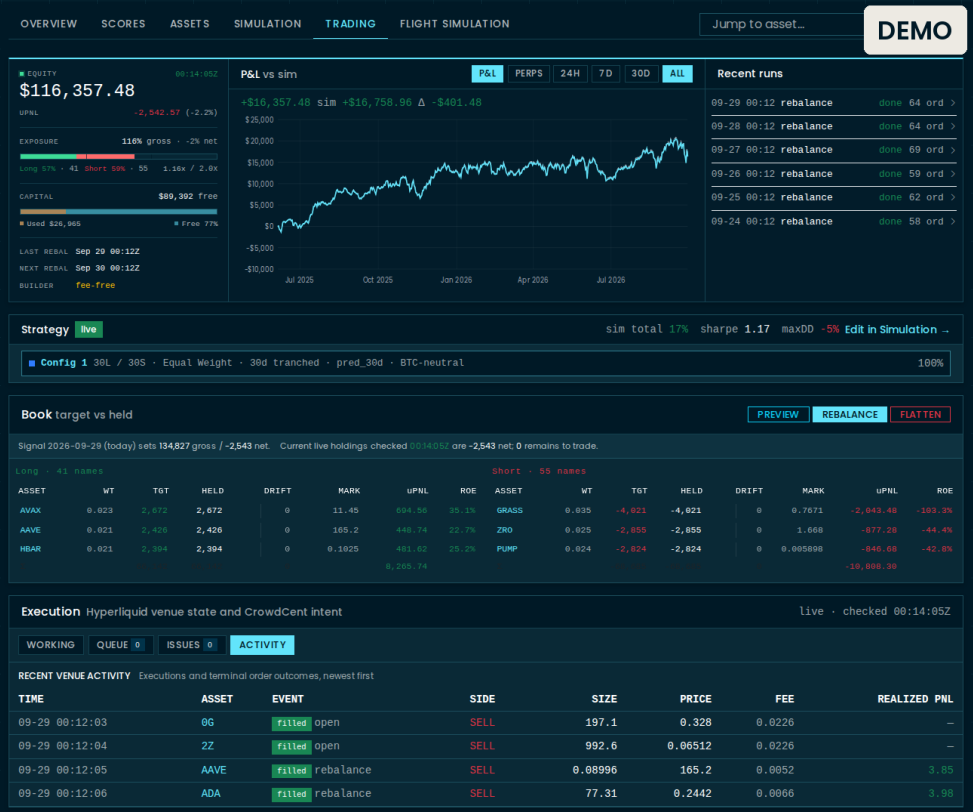
How much of the return came from factor bets, and how much is residual alpha from the signal itself.

Trade the live portfolio directly on Hyperliquid. Just not from here :(

Use your own wallet (self-custody, your keys), preview the trades, fees and turnover, then execute. Automatically schedule rebalances daily.

The US is on the blocked list, so today we can only simulate, but at least that's where crashing is free.

Live trading, where offered, is self-custody on Hyperliquid and is not available to US persons.



Up next: Earn Your Wings

Applied Machine Learning for Systematic Crypto. Soar through CrowdCent Cloud and make a submission or improve your models

WORKSHOP	Right here, right now?
GET READY	crowdcent.com/cloud
SUBMIT	Complete a submission: t-shirt and pin
DOCS	docs.crowdcent.com

SCAN FOR THE CHALLENGE